

Asymptotic LOCC dense coding with the symmetric four-qubit W state

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Abstract

The distributed dense-coding question posed by Bruß and collaborators asks whether the symmetric four-qubit W state gives a strict advantage when two independent senders each transmit one qubit to a different receiver and the receivers decode using local operations and classical communication (LOCC). We give an affirmative answer in their asymptotic unitary-encoding model. Independent Pauli encodings followed by a complete Bell measurement at one receiver produce a finite classical-quantum multiple-access channel whose entire output is local to the other receiver. Winter’s coding theorem gives an achievable sum-rate supremum of $3/2 + h_2(1/4) = 2.311278\dots$ bits per resource copy. In particular, both senders can transmit $9/8$ bits per copy with vanishing average error. This settles the stated W-state classification question using established measurement and coding ingredients; it does not determine the optimal LOCC capacity or assert a one-copy advantage.

1 The question and operational model

Distributed dense coding was formulated in Ref. [1], where the LOCC usefulness of the four-qubit W state was left open. The question was repeated in the 2005 Oberwolfach report, in Bruß’s contribution on pp. 203–205 [2], and the classification was elaborated in Ref. [3]. We use exactly the symmetric resource

$$|W_4\rangle_{A_1A_2B_1B_2} = \frac{1}{2}(|1000\rangle + |0100\rangle + |0010\rangle + |0001\rangle). \quad (1)$$

Initially sender i owns A_i and receiver i owns B_i . For n copies the senders have independent, uniformly distributed messages M_1, M_2 , separate fixed codebooks, and local unitary encoders. Each sends A_i^n noiselessly to receiver i , costing n qubits per sender. The receivers may perform joint operations within their own laboratories and communicate classically across $L^n : R^n$, where $L = A_1B_1$ and $R = A_2B_2$. No additional shared entanglement, sender message disclosure, feedback to senders, or quantum transmission between receivers is supplied. Local decoder ancillas and predetermined codebooks are allowed.

A rate pair (R_1, R_2) is achievable if separate codebooks with $\log_2 N_i(n)/n \rightarrow R_i$ admit such an LOCC decoder with uniform average error tending to zero. Let $C_{\text{LOCC}}(W_4)$ be the supremum of achievable sum rates in this model. All entropies and rates below use base-two logarithms, and $h_2(t) = -t \log_2 t - (1-t) \log_2 (1-t)$. We also give the bridge to the source’s asymptotic accessible-information formulation. Neither zero error nor maximal-error convergence is required.

Theorem 1. *For the resource and routing above, every nonnegative rate pair satisfying*

$$R_1 < \frac{3}{2}, \quad R_2 < \frac{3}{2}, \quad R_1 + R_2 < \frac{3}{2} + h_2(1/4) \quad (2)$$

is achievable. Consequently,

$$C_{\text{LOCC}}(W_4) \geq \frac{3}{2} + h_2(1/4) > 2. \quad (3)$$

The pair $(9/8, 9/8)$ is strictly interior. In the local-unitary classification of Refs. [1, 3], W_4 is therefore LOCC-DC.

2 A Bell instrument with a local quantum output

Let $|\psi^+\rangle = (|01\rangle + |10\rangle)/\sqrt{2}$. In the receiver ordering $L : R$, Eq. (1) is

$$|W_4\rangle = \frac{|\psi^+\rangle_L |00\rangle_R + |00\rangle_L |\psi^+\rangle_R}{\sqrt{2}}. \quad (4)$$

This paired Bell decomposition appears in Cao–Song’s secure-communication protocol [4]; Bell forwarding in a W-state dense-coding setting is also present in Pradhan–Agrawal–Pati [5]. The rate improvement here uses a retained quantum output at the second receiver and asymptotic independent-message coding.

The sender alphabets are $\mathcal{P} = \{I, X, Z, XZ\}$, independently uniform. For letters x, y , sender 1 applies U_x to A_1 and sender 2 applies U_y to A_2 . Receiver 1 measures L in the complete Bell basis $\{\Phi^+, \Phi^-, \Psi^+, \Psi^-\}$, where $|\Phi^\pm\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}$ and $|\Psi^\pm\rangle = (|01\rangle \pm |10\rangle)/\sqrt{2}$. Every outcome j is forwarded to receiver 2; no branch is discarded. Before encoding, the unnormalized residual vectors on R are

$$v_{\Phi^+} = v_{\Phi^-} = \frac{1}{2}|\psi^+\rangle, \quad v_{\Psi^+} = \frac{1}{\sqrt{2}}|00\rangle, \quad v_{\Psi^-} = 0. \quad (5)$$

A Pauli on A_1 permutes the Bell projectors. Writing its induced permutation as π_x (irrelevant branch phases disappear), the normalized output is

$$\omega_{xy}^{JR} = \sum_j |j\rangle\langle j|_J \otimes (U_y \otimes I) |v_{\pi_x(j)}\rangle \langle v_{\pi_x(j)}| (U_y^\dagger \otimes I). \quad (6)$$

The classical register J and both qubits of R are now at receiver 2. Thus $(x, y) \mapsto \omega_{xy}$ is a finite memoryless classical–quantum multiple-access channel with a single local output laboratory. Applied copywise, it respects the original receiver cut.

3 Exact channel entropies

Write $\bar{\omega}_x = \frac{1}{4} \sum_y \omega_{xy}$, $\bar{\omega}_y = \frac{1}{4} \sum_x \omega_{xy}$, and $\bar{\omega} = \frac{1}{16} \sum_{x,y} \omega_{xy}$. The spectra, including multiplicities of positive eigenvalues, are

State	Positive eigenvalues	Entropy
ω_{xy}	$1/2, 1/4, 1/4$	$3/2$
$\bar{\omega}_x$	$1/4$ twice, $1/16$ eight times	3
$\bar{\omega}_y$	$1/8$ eight times	3
$\bar{\omega}$	$3/32$ eight times, $1/32$ eight times	$3 + h_2(1/4)$

Here each output space has dimension 16, and all unlisted eigenvalues are zero. For completeness, these spectra follow directly from Eq. (5). Each conditional Bell block is rank one, with weights $1/4, 1/4, 1/2, 0$. Averaging the A_2 Paulis maps an operator T on $A_2 B_2$ to $I_{A_2}/2 \otimes \text{Tr}_{A_2} T$. The two Φ blocks become $I_4/16$; the Ψ^+ block becomes $I_{A_2}/4 \otimes |0\rangle\langle 0|_{B_2}$. This proves the $\bar{\omega}_x$ row, up to the permutation π_x .

Averaging x distributes each residual branch equally among the four classical labels. Each J block of $\bar{\omega}_y$ is therefore $(U_y \otimes I)\rho_R(U_y^\dagger \otimes I)/4$, with $\rho_R = (|00\rangle\langle 00| + |\psi^+\rangle\langle \psi^+|)/2$, whose nonzero eigenvalues are $1/2, 1/2$. Finally $\rho_{B_2} = \text{diag}(3/4, 1/4)$, so every J block of $\bar{\omega}$ is $I_{A_2}/8 \otimes \rho_{B_2}$, namely $\text{diag}(3, 1, 3, 1)/32$ in the $A_2 B_2$ basis. Its entropy is $5 - (3/4)\log_2 3 = 3 + h_2(1/4)$.

For independent uniform inputs, the conditional Holevo quantities and sum quantity are consequently

$$\begin{aligned}\chi(X : RJ | Y) &= \frac{1}{4} \sum_y S(\bar{\omega}_y) - \frac{1}{16} \sum_{x,y} S(\omega_{xy}) = \frac{3}{2}, \\ \chi(Y : RJ | X) &= \frac{3}{2}, \\ \chi(XY : RJ) &= \frac{3}{2} + h_2(1/4).\end{aligned}\tag{7}$$

4 Coding and implementation of the decoder

Winter's finite classical-quantum multiple-access coding theorem [6], Theorem 9 of the cited preprint, states that for a product input distribution, rates strictly below the individual conditional Holevo quantities and the joint Holevo quantity admit separate deterministic codebooks and a joint output POVM with vanishing uniform average error. The channel in Eq. (6) has finite alphabets, dimension 16, and product priors, and Eq. (7) gives all three constraints. This is an application of that established theorem, rather than a new multiple-access coding theorem. General multiple-access dense-coding applications already appear in Huang-Zhang-Hou [7].

The resulting sender encoders are simply products of the single-qubit Pauli letters in their respective codewords. To see explicitly that the theorem's decoder is permitted here, let $D_{m_1 m_2}$ be its POVM on $J^n R^n$. Every channel output is diagonal in J^n . Replacing each POVM element by

$$D'_{m_1 m_2} = \sum_{\mathbf{j}} (|\mathbf{j}\rangle\langle \mathbf{j}|_{J^n} \otimes I) D_{m_1 m_2} (|\mathbf{j}\rangle\langle \mathbf{j}|_{J^n} \otimes I)$$

preserves positivity, completeness, and every decoding probability. For a received classical string $\mathbf{j}$, its diagonal block is a POVM on R^n , wholly within receiver 2's laboratory. Receiver 1 performs the Bell measurements and forwards $\mathbf{j}$; receiver 2 performs this conditional POVM and may report the decoded labels classically to receiver 1. Only receiver-to-receiver classical communication is used.

For $N_1 N_2 \geq 2$, if the decoder has average error e_n for the uniform message pair $M = (M_1, M_2)$, Fano's inequality gives

$$I(M; \widehat{M}) \geq \log_2(N_1 N_2) - h_2(e_n) - e_n \log_2(N_1 N_2 - 1).\tag{8}$$

When $N_1 N_2 = 1$, the sole message pair is decoded without error and carries zero information, so this case is handled directly. For any fixed positive sum rate, $e_n \rightarrow 0$ makes this information per copy approach $R_1 + R_2$, verifying the accessible-information interpretation as well. Taking rate pairs in Eq. (2) approaching its sum face proves the supremum lower bound in Eq. (3). In particular,

$$h_2(1/4) = 2 - \frac{3}{4} \log_2 3 > \frac{3}{4} \iff 27 < 32,$$

so $(9/8, 9/8)$ is strictly interior and has sum $9/4 > 2$.

To identify the source’s classification shell, each one-qubit marginal of W_4 has eigenvalues $3/4, 1/4$, and each two-qubit marginal has eigenvalues $1/2, 1/2, 0, 0$. The local-unitary LO expression in Refs. [1, 3] is therefore

$$[1 + S(B_1) - S(A_1B_1)] + [1 + S(B_2) - S(A_2B_2)] = 2h_2(1/4) < 2.$$

Allowing the source’s classical fallback gives the comparison value two, without a strict LO advantage. The strict achieved LOCC advantage establishes LOCC-DC in that convention. This completes the proof of Theorem 1.

5 Scope, attribution, and verification

The contribution is the specific achieved lower bound and W-state classification for the original split-receiver asymptotic model. The old Bell decomposition [4], Bell forwarding [5], and multiple-access framework [7, 6] are credited ingredients. The one-copy two-bit Bell-to-Bell construction of Ref. [5], Sec. 4.2.3, is not a converse for the present block protocol. For the uniform 16-letter product ensemble here, measuring both receiver pairs in the Bell basis produces mutual information exactly two bits: its conditional outcomes are four equiprobable pairs, while its average has 16 equiprobable pairs. That calculation concerns this particular measurement only; the second receiver’s quantum output is retained in our coding construction.

Nearby work on the same receiver topology derives LOCC capacity upper bounds [8, 9]; an upper bound above two is not an achieved W-state rate. This comparison refers to the inspected arXiv versions specified in those references. Hayashi–Wang [10] studies a different sender/receiver ownership and routing model; its direct formula cannot be transplanted across the present $A_1B_1 : A_2B_2$ cut. Our literature audit found no earlier explicit or established-equivalent achievability resolution of this exact target. The supplement records the inspected corpus, attribution, and access limitations. This bounded audit does not certify categorical worldwide firstness or that the problem remained open in every subsequent source.

The result is asymptotic with uniform average error. It gives neither an exact optimal capacity, a one-copy advantage, a zero-error or maximal-error code, nor a constructive finite-blocklength error bound. The coding existence theorem is essential; the finite calculations alone do not prove achievability. The accompanying standard-library Python verifier reconstructs all 64 channel blocks from the four-qubit basis using rational arithmetic, checks their spectra by power sums through order 16, and verifies the displayed marginals, entropy coefficients, strict inequality, and both-Bell comparison. Its 551 checks certify the finite algebra, with floating-point values used only for display. The paper’s coding and operational arguments remain analytical.

AI assistance and review status. AI tools were used extensively in developing the solution, drafting the manuscript, reproducing its calculations, and conducting adversarial reviews. These AI reviews do not constitute conventional human peer review or refereeing. This research note is an unrefereed preprint.

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